Sequences — problem 9.45

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The positive real numbers $x_0, \ldots, x_{1995}$ satisfy $x_0 = x_{1995}$ and

$$x_{i-1} + \frac{2}{x_{i-1}} = 2x_i + \frac{1}{x_i}$$

for $i = 1, \ldots, 1995$. Find the maximum value that x_0 can have (IMO 1995).